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In [2]: def new_raphson(f , deriv_f, initial_guess, tolerance=1e-6, max_iteration=100):
        x = initial_guess
        for i in range(max_iteration):
            f_x = f(x)
            deriv_x = deriv_f(x)

            if deriv_x == 0:
                print("Derivative is zero. Newton-Raphson method cannot proceed.")
                return None

            x_new = x - f_x / deriv_x

            if abs(x_new - x) < tolerance:
                return x_new

            x = x_new

        Print("Maximum iteration reached. Method may not have converged.")
        return x
```

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In [ ]:
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